Curriculum Vitae

Personal details

Name:	Shahid Hasnain, PhD
Nationality:	Pakistani
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Professional profile

Dr Shahid Hasnain is a numerical analyst working on the construction, analysis and implementation of high-order schemes for *nonlinear coupled* partial differential equations. His central research line is the numerical treatment of coupled reaction-diffusion and convection-diffusion-reaction systems: the design of high-order finite-difference and finite-volume discretizations, their stability and convergence analysis, and their verification against exact and manufactured solutions. He has applied this methodology to pattern-forming and tumor-growth models, and, in collaborative work, to problems in bio-fluid and thermal transport. He led an HEC-funded project as sole Principal Investigator on the numerical solution of the two-dimensional Gray-Scott coupled nonlinear system, and his first-authored study of finite-difference schemes for the Fisher-KPP equation is his most-cited work.

Education

09/01/2015 - 16/10/2017 - **PhD, Mathematics.** King Abdulaziz University, Jeddah, Saudi Arabia, Faculty of Science. Thesis: *Numerical Study of Reaction-Diffusion Systems*. Supervisor: Prof. Daoud S. Mashat. Cumulative GPA 4.59/5.00.

01/09/2008 - 19/12/2011 - **MS, Computational Sciences (Applied Mathematics).** Linköping University, Sweden. Thesis: *Steady Periodic Water Waves in a Channel of Finite Depth* (asymptotic approach), 123 ECTS.

01/09/2005 - 30/06/2007 - **MSc, Mathematics.** Quaid-i-Azam University, Islamabad, Pakistan.

Professional experience

26/09/2022 - present **Visiting Assistant Professor**, University of Chakwal, Department of Mathematics, Pakistan. Teaching in applied mathematics and numerical analysis.

01/09/2020 - 30/03/2021 - **Assistant Professor (Tenure Track)**, Air University, Multan Campus, Department of Mathematics, Pakistan. Undergraduate teaching in mathematics.

13/06/2018 - 30/06/2019 - **Assistant Professor**, Pakistan Institute of Engineering & Applied Sciences (PIEAS), Islamabad, under the HEC Interim Placement of Fresh PhDs (IPFP) programme. Teaching and research; **Principal Investigator** of the HEC Start-up Research Grant project.

01/01/2013 - 31/10/2013 - **Research Assistant Professor**, Muhammad Ali Jinnah University (now CUST), Islamabad, Pakistan. Research in applied and computational mathematics; contributed to the nationally funded project *Simulation and Control of the Underground Coal Gasification Process* (National ICT R&D Fund), a coupled reactive-transport problem.

Research funding

- **Principal Investigator, HEC Start-up Research Grant Programme (SRGP)**, *Optimized study of the two-dimensional Gray-Scott coupled nonlinear PDE system, with application to tumor-growth modeling*. PIEAS, 2018-2019; final report accepted April 2019. Published outcome: AIP Advances **9** (2019) 105023.

Mentoring and research guidance

12/2022 – 12/2024. At the invitation of his former doctoral supervisor (Prof. D. S. Mashat, King Abdulaziz University), and separately from his teaching appointment at the University of Chakwal, Dr Hasnain provided day-to-day research guidance to a doctoral researcher, Dr Nawal Odah Al-Atawi, through regular online working sessions. The arrangement was informal and carried no institutional title.

Teaching

Undergraduate and graduate teaching in applied mathematics, numerical analysis and nonlinear partial differential equations at the University of Chakwal, Air University, PIEAS and Muhammad Ali Jinnah University (CUST). Professional development: teaching certification (CPDC, University of Chakwal, 2023); English for Specific Purposes (Air University, 2021).

Service to the research community

Peer reviewer for *Archives of Computational Methods in Engineering*, *Discover Nano*, *Physics of Fluids*, *Chaos*, *Case Studies in Thermal Engineering* and *AIP Advances*.

Skills

Numerical methods:	High-order finite-difference and finite-volume schemes; ADI, compact and staggered-grid schemes; stability (von Neumann), consistency and convergence analysis; verification by manufactured solutions and grid-convergence studies.
Programming:	Python, MATLAB, C++; LaTeX.
Simulation:	ANSYS, FLUENT, OpenFOAM; in-house solvers.
Languages:	Urdu (native); English (full professional proficiency); Arabic (working knowledge).

Publications

A full list of publications, along with citation metrics, is available on my Google Scholar profile.

1. **S. Hasnain**, N. O. Al-Atawi, M. Saqib. *Numerical study of magneto-nanolubricant transport through a porous wavy channel using three non-Newtonian rheological models.* Scientific Reports (2026). https://doi.org/10.1038/s41598-026-71090-4
2. B. S. Alshammari, **S. Hasnain**, M. Saqib. *Heat and Mass Transfer in Gold Nanoparticle-Enhanced Synovial Fluid for Biomedical Applications: A Carreau-Yasuda Study.* AIMS Mathematics 11(7) (2026) 22543–22575. https://doi.org/10.3934/math.2026911
3. **S. Hasnain**, M. Saqib, I. Abbas. *Innovative Modeling of Joint Dynamics and Pain Pathways: Numerical Study.* International Communications in Heat and Mass Transfer 169 (2025) 109454. https://doi.org/10.1016/j.icheatmasstransfer.2025.109454
4. **S. Hasnain**, N. O. Al-Atawi, M. Saqib. *Energy and flow conjuncture in knee joint synovial fluid: A Brinkman-based numerical study.* International Journal of Thermal Sciences (2025) 109934. https://doi.org/10.1016/j.ijthermalsci.2025.109934
5. M. Saqib, M. Saleem, M. Abdullah, **S. Hasnain**, M. Farman, M. Hafez. *Exploring the Dynamics of a Generalized Burgers–Fisher Model with Advection and Diffusion Utilizing Finite Volume and Finite Difference Strategies.* AIP Advances 15(11) (2025) 115228. https://doi.org/10.1063/5.0290978
6. M. Saleem, M. Saqib, B. S. Alshammari, **S. Hasnain**, A. Ayesha. *Finite Volume Modeling of Neural Communication: Exploring Electrical Signaling in Biological Systems.* Partial Differential Equations in Applied Mathematics 13 (2025) 101082. https://doi.org/10.1016/j.padiff.2025.101082
7. N. O. Al-Atawi, **S. Hasnain**, M. Saqib. *Parametric Influences on Flow, Heat Transfer, and Nutrient Transport in the Knee Joint Cavity, a Numerical Approach.* Results in Engineering 24 (2024) 103316. https://doi.org/10.1016/j.rineng.2024.103316
8. M. Saleem, M. Saqib, T. O. Alade, Y. Elmasry, M. Farman, **S. Hasnain**. *Advanced Tumor Growth Modeling: A Numerical Study Integrating Phase Plane Analysis with Finite Volume Method.* Ain Shams Engineering Journal 15(12) (2024) 103054. https://doi.org/10.1016/j.asej.2024.103054
9. M. Saleem, T. O. Alade, M. Saqib, **S. Hasnain**, B. S. Alshammari, M. Farman. *Numerical Analysis of Coupled Dynamical Biological Networks: Modeling Electrical Information Exchange among Nerve Cells Using Finite Volume Method.* AIP Advances 14(8) (2024) 085315. https://doi.org/10.1063/5.0215819
10. M. Saqib, F. Akhtar, **S. Hasnain**, I. Abbas, S. Askar, M. Farman. *Dynamical Behavior of Nonlinear Coupled Reaction-Diffusion Model: A Numerical Study Utilizing ADI and Staggered Grid Finite Volume Method in Matlab.* IEEE Access 12 (2024) 12370–12389.

https://doi.org/10.1109/ACCESS.2024.3354812

11. **S. Hasnain**, I. Abbas, N. O. Al-Atawi, M. Saqib, M. F. Afzaal, D. S. Mashat. *Knee Synovial Fluid Flow and Heat Transfer, a Power Law Model.* Scientific Reports 13 (2023) 18184. https://doi.org/10.1038/s41598-023-44482-z
12. I. Abbas, **S. Hasnain**, M. Saqib, D. S. Mashat. *Computational Study on the Influence of Non-Newtonian Nanofluids in Fluid Flow and Heat Transfer over a Permeable Surface with Injection and Suction.* Scientia Iranica (2023), in press. https://doi.org/10.24200/sci.2023.62190.7704
13. N. O. Al-Atawi, **S. Hasnain**, M. Saqib, D. S. Mashat. *Significance of Brinkman and Stokes System Conjuncture in Human Knee Joint.* Scientific Reports 12 (2022) 18992. https://doi.org/10.1038/s41598-022-23402-7
14. M. Saqib, **S. Hasnain**, A. Khaliq, U. Ahmed, N. O. Al-Atawi, D. S. Mashat. *Novel 3D Coupled Convection–Diffusion Model Algorithm.* AIP Advances 12 (2022) 105324. https://doi.org/10.1063/5.0112488
15. I. Abbas, **S. Hasnain**, N. A. Alatawi, M. Saqib, D. S. Mashat. *Thermal Radiation Energy Performance on Stagnation-Point Flow in the Presence of Base Fluids Ethylene Glycol and Water over Stretching Sheet with Slip Boundary Condition.* Energies 15(21) (2022) 7965. https://doi.org/10.3390/en15217965